

**CARMAN, MB, Canada**

WMO: 711470

Lat: **49.500N**Lon: **98.030W**Elev: **880**StdP: **14.23**Time Zone: **-6.00 (NAC)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -23.3      | -18.5 | -29.5                         | 1.1 | -23.1 | -24.6 | 1.5 | -18.1 | 24.9                  | 13.1 | 22.8 | 12.1 | 5.9                      | 300  | 0.666 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.4                   | 86.6            | 70.8 | 83.5 | 69.2 | 80.7 | 67.2 | 75.2                | 82.4 | 72.4 | 79.6 | 70.0 | 77.2 | 11.4                 | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.0                            | 126.5 | 79.7 | 69.9 | 113.7 | 77.1 | 67.4 | 104.0 | 74.2 | 39.3          | 82.3 | 36.7 | 79.7 | 34.6 | 77.5 | 87.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 23.2          | 20.6 | 18.6 | -31.1                      | 93.2 | 5.7                | 2.8 | -35.3                                              | 95.2 | -38.6      | 96.9 | -41.8      | 98.5 | -46.0      | 100.5 |    |    |
| (5)               |      |      | -30.7                      | 79.3 | 5.6                | 3.1 | -34.7                                              | 81.5 | -37.9      | 83.4 | -41.1      | 85.1 | -45.1      | 87.3  |    |    |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 38.0   | 4.8   | 8.6   | 21.6  | 39.4  | 51.9  | 62.4  | 67.2  | 65.3  | 56.3  | 42.0  | 25.0  | 10.3  |  |  |
|                                                                        | DBStd       | 24.60  | 14.79 | 13.64 | 13.62 | 10.82 | 8.65  | 6.37  | 5.25  | 5.74  | 8.11  | 8.77  | 12.05 | 13.87 |  |  |
|                                                                        | HDD50       | 6161   | 1402  | 1158  | 882   | 342   | 82    | 2     | 0     | 0     | 31    | 279   | 751   | 1232  |  |  |
|                                                                        | HDD65       | 10087  | 1867  | 1578  | 1345  | 769   | 412   | 121   | 38    | 66    | 278   | 715   | 1201  | 1697  |  |  |
|                                                                        | CDD50       | 1796   | 0     | 0     | 1     | 23    | 142   | 373   | 532   | 476   | 219   | 30    | 0     | 0     |  |  |
|                                                                        | CDD65       | 247    | 0     | 0     | 0     | 0     | 7     | 42    | 105   | 77    | 16    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 2940   | 0     | 0     | 0     | 9     | 177   | 490   | 1042  | 943   | 263   | 16    | 0     | 0     |  |  |
|                                                                        | CDH80       | 766    | 0     | 0     | 0     | 0     | 45    | 109   | 254   | 290   | 65    | 3     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 9.0    | 9.3   | 9.3   | 9.5   | 10.0  | 10.1  | 8.8   | 7.3   | 7.5   | 8.5   | 9.5   | 9.2   | 9.1   |  |  |
|                                                                        | PrecAvg     | 20.6   | 0.7   | 0.6   | 1.0   | 1.2   | 3.0   | 3.6   | 2.8   | 2.4   | 1.8   | 1.5   | 1.0   | 1.0   |  |  |
|                                                                        | PrecMax     | 26.5   | 1.5   | 2.4   | 2.6   | 3.0   | 5.8   | 7.1   | 6.3   | 5.3   | 3.7   | 6.2   | 3.1   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecMin     | 14.3   | 0.2   | 0.0   | 0.2   | 0.2   | 1.0   | 1.3   | 0.6   | 0.6   | 0.1   | 0.2   | 0.1   | 0.2   |  |  |
|                                                                        | PrecStd     | 3.5    | 0.4   | 0.5   | 0.7   | 0.7   | 1.3   | 1.5   | 1.4   | 1.3   | 0.8   | 1.3   | 0.8   | 0.5   |  |  |
|                                                                        | DB          | 41.0   | 41.7  | 58.8  | 74.8  | 86.1  | 87.5  | 89.4  | 91.2  | 86.5  | 75.8  | 58.5  | 42.1  |       |  |  |
|                                                                        | MCWB        | 37.4   | 37.5  | 48.5  | 55.3  | 61.5  | 72.0  | 76.5  | 72.0  | 66.8  | 59.8  | 49.4  | 37.3  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 36.5   | 36.9  | 49.5  | 68.2  | 78.7  | 83.1  | 85.6  | 86.9  | 81.0  | 67.6  | 51.0  | 36.9  |       |  |  |
|                                                                        | 2% MCWB     | 33.7   | 34.4  | 43.8  | 52.3  | 58.2  | 66.9  | 73.5  | 70.2  | 64.6  | 55.1  | 44.0  | 34.2  |       |  |  |
|                                                                        | 5% DB       | 31.6   | 33.3  | 42.9  | 63.1  | 73.9  | 79.5  | 82.7  | 83.0  | 76.4  | 61.8  | 45.5  | 32.3  |       |  |  |
|                                                                        | 5% MCWB     | 30.2   | 32.1  | 39.2  | 50.2  | 57.1  | 65.5  | 71.1  | 68.8  | 62.9  | 52.5  | 40.8  | 30.7  |       |  |  |
|                                                                        | 10% DB      | 26.4   | 29.4  | 37.9  | 57.5  | 68.9  | 76.2  | 79.9  | 79.5  | 71.3  | 56.4  | 40.4  | 29.0  |       |  |  |
|                                                                        | 10% MCWB    | 25.1   | 28.1  | 35.5  | 47.8  | 55.3  | 64.3  | 69.2  | 66.9  | 60.1  | 49.5  | 37.5  | 27.7  |       |  |  |
|                                                                        | 0.4% WB     | 37.5   | 38.3  | 51.2  | 59.0  | 66.5  | 77.0  | 80.2  | 77.0  | 72.0  | 61.7  | 50.1  | 37.8  |       |  |  |
|                                                                        | 0.4% MCDB   | 41.0   | 40.9  | 56.8  | 67.3  | 76.7  | 83.1  | 85.9  | 85.7  | 81.3  | 72.0  | 57.2  | 41.2  |       |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 34.0   | 34.6  | 44.5  | 54.7  | 62.5  | 72.0  | 76.3  | 73.4  | 67.5  | 57.0  | 44.9  | 34.4  |       |  |  |
|                                                                        | 2% MCDB     | 36.4   | 36.5  | 50.5  | 65.2  | 71.9  | 77.8  | 82.7  | 82.3  | 75.0  | 65.5  | 50.1  | 36.9  |       |  |  |
|                                                                        | 5% WB       | 30.2   | 32.3  | 39.6  | 51.3  | 59.7  | 68.4  | 73.5  | 70.8  | 64.8  | 53.3  | 40.9  | 31.0  |       |  |  |
|                                                                        | 5% MCDB     | 31.6   | 33.6  | 43.5  | 61.4  | 68.6  | 75.4  | 79.8  | 79.3  | 73.0  | 60.6  | 44.9  | 32.6  |       |  |  |
|                                                                        | 10% WB      | 25.3   | 28.3  | 35.9  | 48.3  | 57.3  | 65.8  | 70.9  | 68.5  | 61.8  | 49.8  | 37.5  | 27.8  |       |  |  |
|                                                                        | 10% MCDB    | 26.3   | 29.5  | 38.0  | 57.4  | 67.0  | 73.6  | 77.6  | 76.5  | 69.4  | 56.3  | 40.4  | 29.0  |       |  |  |
|                                                                        | MDBR        | 16.9   | 17.7  | 16.8  | 20.1  | 22.3  | 20.1  | 21.4  | 23.5  | 22.3  | 18.3  | 15.6  | 15.2  |       |  |  |
|                                                                        | 5% DB       | 20.2   | 18.2  | 20.3  | 31.3  | 32.4  | 26.3  | 25.4  | 30.0  | 30.9  | 28.9  | 23.8  | 18.2  |       |  |  |
| (36) Mean Daily Temperature Range                                      | MCWBR       | 18.4   | 15.8  | 15.0  | 17.7  | 16.3  | 13.7  | 15.1  | 15.4  | 16.4  | 17.5  | 17.1  | 16.2  |       |  |  |
|                                                                        | MCDBR       | 20.4   | 17.5  | 20.3  | 28.5  | 25.0  | 22.1  | 23.5  | 25.6  | 26.5  | 27.2  | 22.6  | 18.1  |       |  |  |
|                                                                        | MCWBR       | 18.6   | 15.2  | 15.3  | 17.4  | 14.6  | 13.9  | 15.7  | 15.1  | 15.8  | 17.1  | 16.7  | 16.1  |       |  |  |
|                                                                        | MCWBR       | 18.6   | 15.2  | 15.3  | 17.4  | 14.6  | 13.9  | 15.7  | 15.1  | 15.8  | 17.1  | 16.7  | 16.1  |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.270  | 0.292 | 0.302 | 0.358 | 0.374 | 0.380 | 0.388 | 0.383 | 0.348 | 0.304 | 0.271 | 0.251 |       |  |  |
|                                                                        | taud        | 2.257  | 2.214 | 2.359 | 2.329 | 2.344 | 2.355 | 2.341 | 2.347 | 2.436 | 2.536 | 2.480 | 2.375 |       |  |  |
|                                                                        | Ebn at Noon | 247    | 269   | 289   | 282   | 280   | 279   | 274   | 270   | 270   | 267   | 252   | 243   |       |  |  |
|                                                                        | Edh at Noon | 23     | 30    | 32    | 36    | 38    | 38    | 37    | 36    | 30    | 22    | 18    | 17    |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 460    | 843   | 1243  | 1584  | 1746  | 1878  | 1991  | 1633  | 1190  | 720   | 405   | 327   |       |  |  |
|                                                                        | RadStd      | 43     | 63    | 103   | 117   | 139   | 134   | 81    | 107   | 87    | 96    | 56    | 39    |       |  |  |

**Historical Trends**

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | -1.64 | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (28 neighbors) | N/A     | N/A    | N/A     | N/A   | -1.30 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page